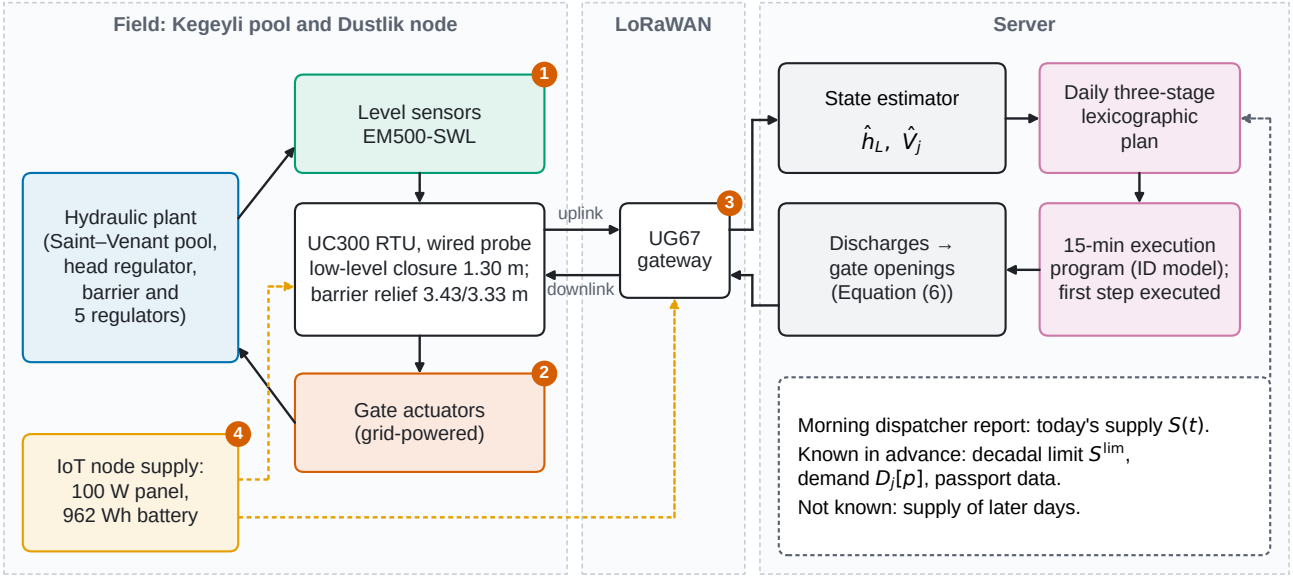
IoT constraints:
1 sensor error
2 quantisation, deadband
3 packet budget, packet loss
4 node energy reserve
Field: Kegeyli pool and Dustlik node
LoRaWAN
Server
Level sensors EM500-SWL
Hydraulic plant (Saint–Venant pool, head regulator, barrier and 5 regulators)
UC300 RTU, wired probe low-level closure 1.30 m; barrier relief 3.43/3.33 m
uplink
downlink
UG67 gateway
Gate actuators (grid-powered)
IoT node supply: 100 W panel, 962 Wh battery
State estimator
$\hat{h}_L,\ \hat{V}_j$
Daily three-stage lexicographic plan
Discharges → gate openings (Equation (6))
15-min execution program (ID model); first step executed
Morning dispatcher report: today's supply $S(t)$.
Known in advance: decadal limit S^{lim}, demand $D_j[p]$, passport data.
Not known: supply of later days.